

Riparian Resolution Comparison (25m vs 30m)

```
library(SpaDES.core)
library(SpaDES.project)
library(terra)
library(sf)
library(dplyr)

root <- "E:/RiparianBuffers"

setPaths(
  modulePath = file.path(root, "modules"),
  inputPath  = file.path(root, "inputs"),
  outputPath  = file.path(root, "outputs"),
  cachePath  = file.path(root, "cache"),
  scratchPath = file.path(root, "scratch")
)

getModule(
  "shirinvark/RiparianBuffers",
  modulePath = getPaths()$modulePath,
  overwrite  = TRUE
)
```

```
## $success
## [1] "shirinvark/RiparianBuffers@HEAD"
##
## $failed
## character(0)
```

Data Preparation

```
rivers_zip <- "D:/HydroRIVERS_v10_na_shp (3).zip"
rivers_dir <- "D:/HydroRIVERS"
if (!dir.exists(rivers_dir)) unzip(rivers_zip, exdir = rivers_dir)

river_shp <- list.files(rivers_dir, pattern="\\.shp$", recursive=TRUE, full.names=TRUE)
streams_all <- terra::vect(river_shp)
streams_all <- terra::project(streams_all, "EPSG:3857")

lakes_zip <- "D:/HydroLAKES_polys_v10_shp.zip"
lakes_dir <- "D:/HydroLAKES"
if (!dir.exists(lakes_dir)) unzip(lakes_zip, exdir = lakes_dir)

lake_shp <- list.files(lakes_dir, pattern="\\.shp$", recursive=TRUE, full.names=TRUE)
```